

Alwin Joseph

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PROFILE

Software developer specialising in AI-powered web and mobile applications. Experienced in building high-performance React and Flutter interfaces integrated with live AI systems. Strong eye for UI/UX and SEO. Looking to build products with great design and intelligent systems.

TECHNICAL SKILLS

- **Frameworks & Libraries:** React, Node.js, Express, TailwindCSS, Three.js, Flutter
- **Programming:** Python, HTML5, CSS3, JavaScript, TypeScript, C++, C, Dart.
- **AI & ML:** YOLOv8, Pegasus, Computer Vision, Deep Learning, Generative AI, Prompt Engineering
- **Tools & Version Control:** Git, GitHub, Figma

PROJECTS

ELIS-Explain like I'm Stuck - Diagnostic AI Tutor - *Developer*

- Built a Socratic debugging assistant on Gemini API that identifies logical gaps in code without giving direct answers.
- Engineered a constrained multi-turn prompt architecture to prevent the model from slipping into answer mode even under direct user pressure.
- Built a Node.js/Express backend to proxy Gemini API requests, keeping API keys server-side and enabling request rate-limiting.
- Implemented persistent chat history so sessions survive page refreshes.
- Stack: React, Node.js, Express, TailwindCSS, Gemini API, Context Modelling.

SHMS - Smart Health Management System - *AI Researcher & Model Developer*

- Built an end-to-end medical imaging pipeline on MIMIC-CXR and NIH Chest X-Ray datasets — two of the largest public medical imaging repositories.
- Trained YOLOv5 to detect and localise disease markers across chest X-rays.
- Integrated Pegasus transformer to auto-generate plain-language diagnostic summaries from medical reports.

Personal Portfolio - *Developer*

- Built a React + TailwindCSS portfolio
- Achieved a Lighthouse Performance Score of 96.
- Optimised for search discoverability through semantic HTML5 structure and custom meta tags.
- Stack: React, TailwindCSS, Three.js — deployed on Netlify.

MotoBuddy - Community Riding Platform | *Flutter Developer*

- Designed and built a cross-platform Flutter app for motorcycle enthusiasts to discover routes, organise group rides, and track milestones.
- Designed and Prototyped screens in Figma.
- Implemented custom micro-interactions and skeletal loading states to minimise perceived latency.

EXPERIENCE

Website Design & Development Intern — Keltron Controls — Aroor 2024

- Engineered the user-to-component communication interface for the Underwater Communication Protocol System (UCAPS) - a live embedded systems product — using HTML, CSS, JavaScript, and jQuery.
- Fetched and rendered real-time hardware state by consuming third-party API calls via jQuery.
- Collaborated on responsive layout design and visual component development across the product interface.

EDUCATION

Rajagiri School of Engineering and Technology, Kakkanad

B.Tech in Computer Science and Business Systems

2021 — 2025